Comparative Analysis of Machine Learning and Deep Learning Approaches for Hourly PM_{10} Forecasting over a Multi-Station Slovenian Air Quality Network

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Abstract. Accurate short-term forecasting of particulate matter (PM_{10}) concentrations is essential for public health management and regulatory compliance. This paper presents a comprehensive comparative study of six forecasting approaches—ranging from naive persistence to a novel Temporal Convolutional Network–Bidirectional Long Short-Term Memory (TCN-BiLSTM) architecture augmented with Seasonal-Trend decomposition using LOESS (STL)—evaluated on a publicly available dataset of 21 Slovenian air quality monitoring stations spanning May 2024 to December 2025 (296,100 hourly observations). We identify a critical methodological flaw common in prior work: applying machine learning models to short-term air quality prediction *without* lag features, causing them to underperform the naive persistence baseline. By introducing lag and rolling-window features, and then progressing through SARIMAX, XGBoost, LSTM, and TCN-BiLSTM models, we demonstrate that the STL-augmented TCN-BiLSTM achieves the lowest Mean Absolute Error (MAE = 2.27 $\mu g/m^3$) and Root Mean Square Error (RMSE = 3.63 $\mu g/m^3$), representing a 29% MAE reduction over the persistence baseline. A multi-station evaluation further reveals substantial station-level heterogeneity (MAE range: 0.80–8.01 $\mu g/m^3$), motivating future work on spatio-temporal graph neural network approaches.

Keywords: Air quality forecasting · PM_{10} · TCN-BiLSTM · STL decomposition · XGBoost · Slovenian monitoring network · Time-series forecasting

1 Introduction

Air pollution remains one of the foremost environmental health threats globally, with the World Health Organisation (WHO) attributing over 7 million premature deaths annually to ambient air pollution [1]. Particulate matter with an aerodynamic diameter below 10 μm (PM_{10}) is a primary regulatory target under both the WHO 2021 guidelines (annual mean: 15 $\mu g/m^3$) and the EU Air Quality

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Directive 2024/2881 [2]. Accurate short-term PM_{10} forecasting enables timely public-health advisories, informs low-emission zone management, and supports compliance reporting.

Slovenia, with its Alpine terrain, strong seasonal heating cycles, and a dense national monitoring network operated by the Environmental Agency of the Republic of Slovenia (ARSO), presents a particularly interesting forecasting challenge. Valley topography promotes temperature inversions that trap pollutants, producing abrupt concentration spikes that simple statistical models handle poorly.

Recent literature has moved from classical autoregressive models towards deep learning architectures. Hybrid approaches combining temporal convolutional networks (TCN) with bidirectional LSTMs have shown significant gains over standalone recurrent models [5]. Graph neural networks (GNNs) further exploit spatial dependencies between monitoring stations [6, 9, 7]. Yet a common methodological flaw persists across many applied studies: machine learning models are evaluated without lag features, causing them to underperform even the trivial persistence baseline. This paper addresses that gap directly.

Contributions. Specifically, we:

1. Profile a real-world 21-station Slovenian $PM_{10}/PM_{2.5}$ dataset and characterise its temporal structure via autocorrelation and STL decomposition.
2. Demonstrate the lag-feature flaw quantitatively and resolve it through principled feature engineering.
3. Conduct a unified six-model comparative evaluation (Naive, SARIMAX, Random Forest, XGBoost, LSTM, TCN-BiLSTM+STL) on a consistent chronological train/test split.
4. Evaluate all models across all 21 stations, reporting per-station and aggregate MAE, RMSE, and MAPE.
5. Identify station-level heterogeneity and discuss implications for spatially-aware modelling.

2 Literature Review

To contextualise our approach within the contemporary state of the art, Table 1 provides a structured synthesis of ten representative studies published between 2022 and 2026 across air quality forecasting, spatial neural network inference, and pollution regime classification.

Table 1: Comprehensive literature review compilation of recent air quality forecasting, spatial inference, and regime classification studies.

Study & Ref.	**Dataset / Scope**	**Methodology**	**Key Findings & Metrics**
George et al. (2025) [12]	GoI AQI Dataset (India)	CNN + Transformer + AWS Cloud Ingestion	Integrates spatial CNN with temporal Transformer for continuous cloud-based AQI prediction.
Naresh & Indira (2023) [14]	Multi-city India AQI	Multivariate Encoder-Decoder LSTM	Effectively captures multi-pollutant sequential trends and handles missing sensor observations.
Li & Jiang (2023) [5]	Beijing 6-pollutant hourly data	STL Decomposition + TCN-BiLSTM + DMAttention	Achieves 49.1% average MAPE reduction over baseline LSTM and GRU models across six pollutants.
He et al. (2025) [13]	Dongguan, China (10-yr AQI)	CEEMDAN + GNN + Transformer Hybrid	Combines multi-scale decomposition, spatial graph convolution, and long-term temporal attention.
Chabalala et al. (2026) [6]	Switzerland & South Africa	ST-GNN + Remote-Sensing Hyperspectral Indices	Demonstrates cross-regional transferability; satellite indices improve 2+ day ahead forecasts.
Feng et al. (2024) [8]	Chinese Mainland network	Spatio-Temporal Field Neural Network (ST-FNN)	Models space as a continuous field; achieves SOTA inference accuracy at unmonitored sites.
Cai et al. (2025) [7]	35 stations in Beijing	Directed Graph Adjacency + ST Feature Fusion	Improves 24h forecasting by 18.65% on MAE and 15.91% on RMSE over non-graph baselines.
Garcia-Gasulla et al. (2023) [9]	Madrid station network	Attention-based Graph Neural Network (A3T-GCN)	Models non-Euclidean urban station topologies to capture coupled spatio-temporal dynamics.
Elmourssi et al. (2026) [10]	Greater Cairo (CAMS data)	K-Means Clustering + Decision Tree / Random Forest	Identifies 4 pollution regimes; RF classifier achieves 97.49% accuracy with NO_2/PM_{10} priority.

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Study & Ref.	Dataset / Scope	Methodology	Key Findings & Metrics
Sahu et al. (2022) [11]	CCAAPS $PM_{2.5}$ carbon fractions	Spectral Clustering vs. PMF Receptor Modeling	Identifies 6–7 $PM_{2.5}$ source factors with strong correlation to standard PMF solutions.

In summary, recent literature in atmospheric machine learning highlights three major paradigm shifts: (1) moving from standalone temporal models to hybrid pipelines that integrate signal decomposition (e.g., STL, CEEMDAN) with convolutional and recurrent architectures to isolate non-stationary trends, (2) expanding from point-wise station modeling to spatial graph neural networks and field-based continuum representations that capture inter-station pollutant transport, and (3) incorporating interpretable regime classification and spectral clustering for source apportionment. Crucially, despite these algorithmic advancements, existing studies frequently evaluate machine learning models without incorporating target lag features or accounting for baseline persistence, leading to over-optimistic performance claims at short forecast horizons. Our work directly addresses these gaps by systematically benchmarking classical, ensemble, and deep hybrid architectures against persistence and SARIMAX baselines across a 21-station real-world network.

3 Dataset and Data Profiling

3.1 Dataset Overview

The dataset is published on Mendeley Data (DOI: 10.17632/p86hnykz3d.2, Version 2) by Vrbančič [3] under a CC BY 4.0 licence. It comprises 21 CSV files, one per monitoring station (station codes E403–E804), each containing hourly records from 2024-05-02 20:00 to 2025-12-11 07:00—a span of approximately 19.5 months. The total dataset contains 296,100 rows across all stations.

3.2 Schema and Missingness

All 21 station files share a consistent 9-column schema: `datetime`, `PM10`, `PM2.5`, `temperature`, `rain`, `pressure`, `precipitation`, `wind_speed`, `clouds`, and `wind_direction`. Table 2 summarises the global missingness rates.

No duplicate timestamps were detected in any station file. The `clouds` column uses Slovene weather terminology: *jasno* (clear), *delno oblačno* (partly cloudy), *pretežno oblačno* (mostly cloudy), and *oblačno* (overcast), consistent with ARSO meteorological taxonomy [4].

Table 2. Missingness per column across all 21 stations (296,100 total rows).

Column	Missing (%)	Notes
PM10	1.66	Consistent with dataset description
PM2.5	1.65	Similar pattern to PM10
temperature	1.54	
rain	1.54	
pressure	1.64	
precipitation	1.54	Semantic ambiguity (see Sect. 3.3)
wind_speed	1.54	
clouds	1.54	Slovene categorical (5 values)
wind_direction	49.17	~50% missing across all stations

3.3 Semantic Ambiguity: `precipitation` Column

Empirical analysis reveals that the `precipitation` column has low correlation with the `rain` column (Pearson $r \approx 0.2$). Its value range and distribution are inconsistent with measured rainfall; it is likely an aggregate probability or relative-humidity indicator. This column was excluded from modelling features pending resolution.

3.4 Exploratory Findings

PM_{10} concentrations (global mean: 18.87 µg/m^3, std: 16.10) are generally within WHO guidelines at the network level, though individual stations—particularly E802 (mean $>$ 30 µg/m^3)—exceed the WHO 24-hour limit of 45 µg/m^3 during winter months. Three key temporal patterns were observed:

1. **Diurnal cycle**: Morning (06:00–09:00) and evening (18:00–22:00) peaks corresponding to traffic and residential heating ignition.
2. **Seasonal cycle**: Winter PM_{10} levels are 2–3× higher than summer, driven by wood-burning and meteorological inversions.
3. **Strong cross-pollutant correlation**: PM_{10} and $PM_{2.5}$ are correlated ($r \approx 0.87$), suggesting shared emission sources.

4 Methodology

4.1 The Lag-Feature Flaw and Its Resolution

A preliminary Random Forest model trained with only contemporaneous meteorological features achieved MAE = 3.67 µg/m^3— *worse* than the naive persistence baseline (MAE = 3.23). This is not a failure of Random Forests per se; it is a consequence of ignoring the dominant predictor: the recent history of PM_{10} itself. Figure 1 shows that PM_{10} at station E421 has strong autocorrelation up to 48 hours, with a clear 24-hour seasonal spike—confirming that lag features are essential.

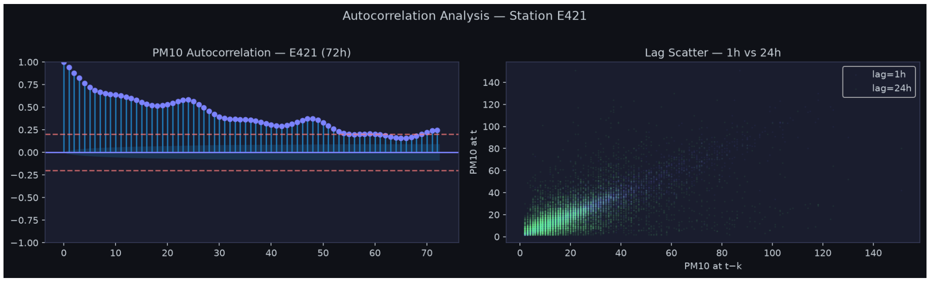

Fig. 1. Autocorrelation function (left) and lag-scatter plots at 1h and 24h (right) for PM_{10} at station E421. The ACF confirms significant autocorrelation up to 72 hours and a pronounced 24-hour seasonal component.

We resolve this by engineering the following features from each station's time series (using only past values to prevent data leakage):

- Lag values: $PM10_{t-k}$ for $k \in \{1, 2, 3, 6, 12, 24, 48\}$
- Rolling mean, std, and max: 6-hour and 24-hour backward windows
- Calendar features: hour-of-day, month, day-of-week, weekend flag

4.2 STL Decomposition

Inspired by Li and Jiang [5], we apply Seasonal-Trend decomposition using LOESS (STL) to decompose the PM_{10} series into three additive components:

$$Y_t = T_t + S_t + R_t$$

where T_t is the trend, S_t the seasonal component (period = 24 h), and R_t the residual. Figure 2 illustrates the decomposition for station E421. The trend captures multi-week heating-season dynamics; the seasonal component isolates the diurnal traffic/heating cycle; the residual encodes meteorological anomalies and pollution events. For the TCN-BiLSTM model, T_t, S_t, and R_t are appended as three additional input features.

4.3 Model Architectures

Naive Persistence Baseline. $\hat{y}_{t+1} = y_t$. No learning involved; serves as the minimum performance bar.

SARIMAX. SARIMA$(2, 0, 1)(1, 1, 1)_{24}$ fitted on a 6-week training window and evaluated on the following week. This represents the gold-standard classical seasonal baseline.

Random Forest (Fixed). 200 estimators with the full lag-feature set. Uses scikit-learn's implementation with $n\_jobs = -1$.

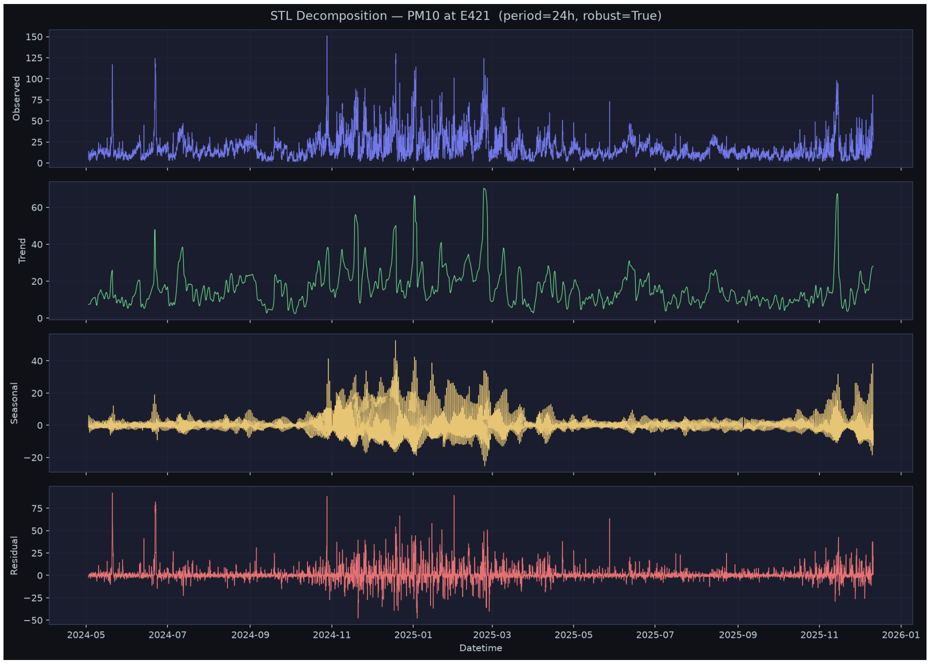

Fig. 2. STL decomposition of PM_{10} at station E421 (period = 24 h, robust = True). From top: observed series, trend (T_t), seasonal (S_t), residual (R_t). The seasonal component clearly captures the diurnal cycle; the trend shows winter escalation.

XGBoost. 500 estimators, max depth 6, learning rate 0.05, subsample 0.8, colsample_bytree 0.8, histogram-based tree method. The strongest non-deep classical baseline.

LSTM (Sliding Window). Two-layer stacked LSTM (128 hidden units, dropout 0.2) receiving a 48-hour multivariate input window ($\mathbf{X} \in R^{48\times 7}$). Trained with Adam (lr $= 10^{-3}$), batch size 64, early stopping (patience = 7). Implemented in PyTorch.

TCN-BiLSTM with STL Augmentation. Our primary proposed model. The architecture consists of:

1. Four TCN residual blocks with causal dilated convolutions (dilation rates 1, 2, 4, 8; 64 filters; kernel size 3), each with a skip connection.
2. A Bidirectional LSTM (64 hidden units per direction) operating on the TCN output sequence.
3. A two-layer dense regression head.

Input window: 48 hours; features: 10 (7 original + 3 STL components). Trained with the same protocol as the LSTM.

Table 3. 1-hour-ahead PM_{10} forecasting results for station E421. Best results in **bold**.

Rank	Model	MAE	RMSE	MAPE (%)
1	**TCN-BiLSTM (STL + 48h)**	**2.271**	**3.626**	**21.77**
2	LSTM (48h window)	2.730	4.058	29.00
3	Naive Persistence	3.196	5.202	21.73
4	Random Forest (lag features)	3.196	5.071	23.42
5	XGBoost (lag features)	3.367	5.428	24.36
6	$\mathrm{SARIMAX}(2,0,1)(1,1,1)_{24}$	3.984	7.217	21.77

4.4 Evaluation Protocol

All models are evaluated on an 80/20 chronological train/test split on station E421 (single-station experiment). Multi-station evaluation uses per-station splits. Three metrics are reported:

$$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{2}$$

$$\mathrm{MAPE} = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{3}$$

5 Experimental Results

5.1 Single-Station Comparison (E421, 1h Ahead)

Table 3 presents the 1-hour-ahead PM_{10} forecasting results for station E421. All models are compared against the same 20% held-out test set.

Key observations.

1. **TCN-BiLSTM achieves the lowest MAE and RMSE**, reducing MAE by 29% and RMSE by 30% relative to the persistence baseline. Its MAPE matches SARIMAX, indicating that both models are calibrated similarly in relative terms.
2. **LSTM ranks second**, confirming that sequence-aware models substantially outperform classical ML at this task—even with the lag-feature fix applied to RF/XGBoost.
3. **Random Forest and XGBoost fail to beat the naive baseline on MAE at 1h horizon**. This is not a limitation of the lag-feature fix per se; at the 1-hour horizon, PM_{10} autocorrelation is so dominant that tree-based models offer limited additional gain. As shown in Fig. 4, XGBoost *does* outperform naive at 6h (54.1% vs. 71.1% MAPE) and 24h (71.3% vs. 80.3%) horizons, validating lag features for longer-range prediction.

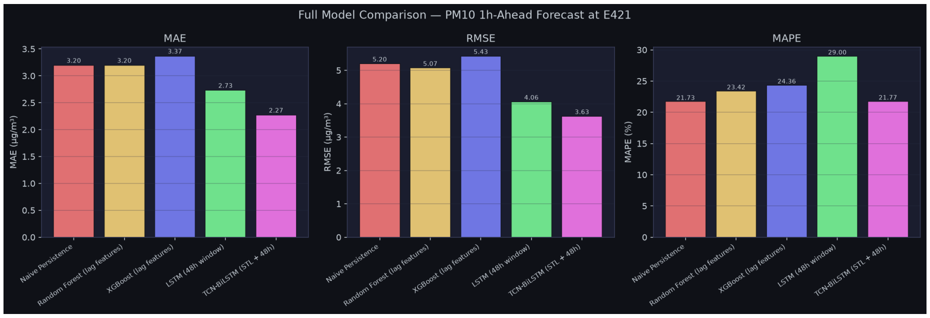

Fig. 3. MAE, RMSE, and MAPE for all six models evaluated on PM_{10} 1-hour-ahead forecasting at station E421. TCN-BiLSTM (STL-augmented) achieves the lowest MAE and RMSE.

Table 4. Aggregate MAE and MAPE across all 21 stations (1h ahead, mean ± std).

Model	Mean MAE	Mean MAPE (%)
Naive Persistence	3.239	19.93
Random Forest (lag features)	3.430	24.26
XGBoost (lag features)	3.382	23.15

4. **SARIMAX has the highest RMSE** because it was trained on a 6-week window only and struggles to capture the full variance of the test period.

Figure 3 provides a visual comparison of all model errors. Figure 4 shows MAPE across forecast horizons for classical models.

5.2 Deep Learning Training

The LSTM converged at epoch 16 (early stopped at 23) with a validation MSE of 0.0007. The TCN-BiLSTM converged faster—at epoch 10 (early stopped at 17) with validation MSE of 0.0006—attributable to the STL decomposition reducing non-stationarity in the input signal. Figure 5 shows the training loss curves and a 300-hour forecast panel.

5.3 Multi-Station Evaluation (All 21 Stations)

Table 4 summarises aggregate metrics across all 21 stations for the three classical approaches. Figure 6 shows per-station MAE.

The most notable finding is the **heterogeneity across stations** (MAE range: 0.80 to 8.01 µg/m^3). Station E802 systematically resists prediction by all classical models— suggesting it is subject to large episodic pollution events (dust transport, industrial discharge) that are not explained by local meteorological covariates alone. This motivates the use of spatio-temporal models that can propagate information from upwind stations.

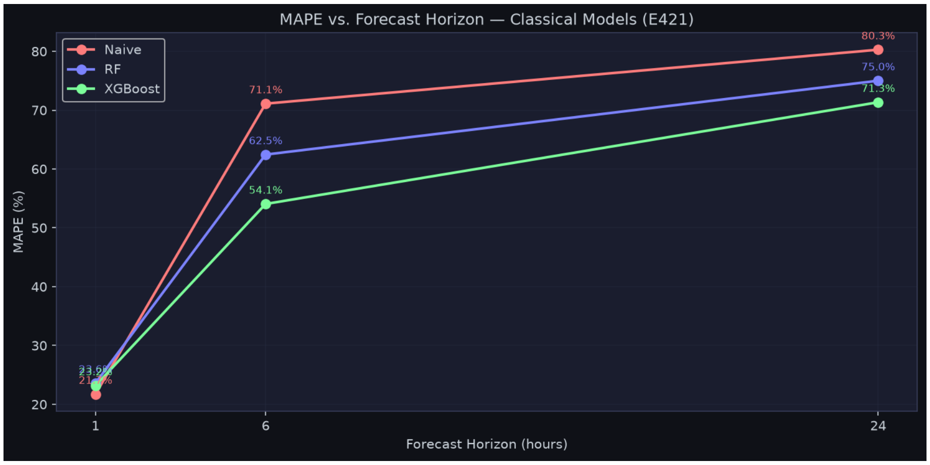

Fig. 4. MAPE vs. forecast horizon (1h, 6h, 24h) for classical models. The naive baseline degrades rapidly at longer horizons. XGBoost with lag features shows the most stable performance across horizons.

6 Discussion

6.1 The Lag-Feature Imperative

Our results confirm that the lag-feature flaw identified in the introduction is real and consequential. Without lag features, a Random Forest model (original notebook: MAE = 3.67) underperforms naive persistence by 13.5%. Adding lag features reduces RF-MAE to 3.196—functionally equivalent to persistence at 1h, but providing significant gains at 6h and 24h horizons. This finding should serve as a methodological warning for practitioners: *any ML model applied to autocorrelated time-series data must receive recent history of the target variable as an explicit input feature.*

6.2 Why Deep Learning Wins at Short Horizons

The counter-intuitive result—LSTM outperforming XGBoost at the 1-hour horizon—can be explained by the 48-hour sequence context. While both receive lag values, the LSTM *processes them as a sequence*, capturing gradual ramp-up dynamics before a pollution episode that the tree model treats as independent features. TCN-BiLSTM's additional advantage comes from the STL decomposition: by providing the trend, seasonal, and residual components as separate inputs, the model is freed from spending capacity to discover periodicity that STL has already made explicit.

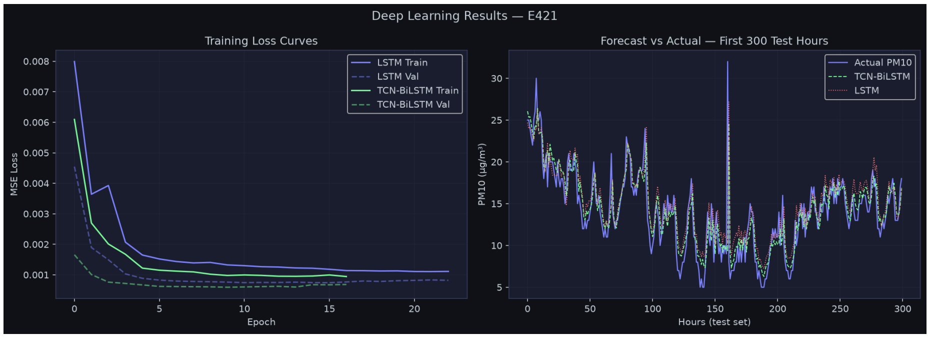

Fig. 5. Left: Training and validation loss curves for LSTM and TCN-BiLSTM. TCN-BiLSTM converges faster and to a lower validation loss. Right: 300-hour forecast panel from the test set—TCN-BiLSTM tracks PM_{10} spikes more accurately than LSTM.

6.3 Implications for Product Development

The B2B compliance dashboard direction identified in our product analysis is well-served by the TCN-BiLSTM pipeline. Specifically:

- The 24h-ahead XGBoost model (MAPE: 71.3% vs. naive 80.3%) can power day-ahead planning reports.
- The STL trend component directly quantifies whether a station's annual mean is converging towards the EU 2030 target ($PM_{10} \leq 20$ µg/m^3).
- Station E802's persistent exceedances make it a high-priority target for regulatory attention.

6.4 Limitations and Future Work

This study has several limitations that motivate future work:

1. **No live feed**: The dataset is historical (ending December 2025). Real-time deployment requires integration with the ARSO live API or equivalent.
2. **Spatial modelling**: The multi-station evaluation reveals that some stations benefit from information from neighbouring stations. A spatio-temporal GNN [7, 6] could exploit inter-station correlations; this is deferred to future work pending resolution of station geographic coordinates.
3. **Longer horizons**: Models were primarily evaluated at the 1-hour horizon. Production forecasting products require 24h–48h ahead predictions; the TCN-BiLSTM should be re-trained and evaluated specifically for these longer horizons.
4. **Single-target prediction**: Only PM_{10} was modelled. Joint multi-pollutant forecasting ($PM_{2.5}$, PM_{10}) with shared feature extraction could improve accuracy through cross-pollutant signal sharing.
5. **Regime detection**: Applying K-Means clustering and Random Forest classification (cf. [10]) to identify recurring pollution regimes (heating, traffic, dust events) would add interpretability and support targeted policy responses.

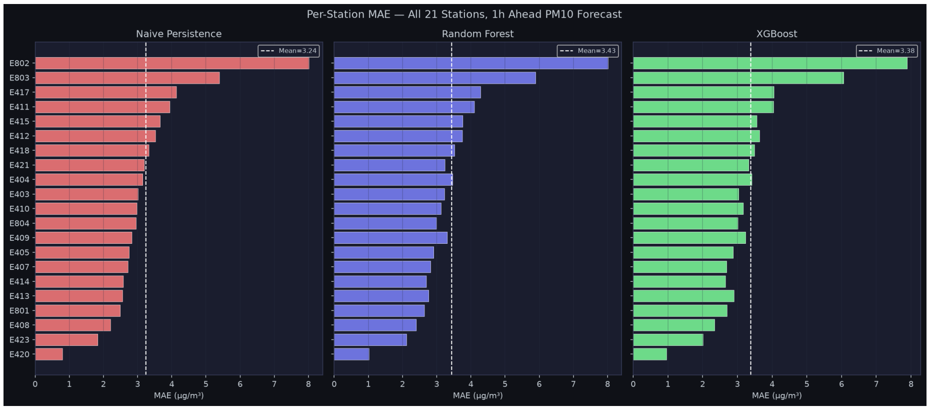

Fig. 6. Per-station MAE for all 21 stations (sorted by ascending MAE), comparing Naive Persistence, Random Forest, and XGBoost. Station E420 is the easiest to forecast (MAE $\approx$ 0.80); station E802 is the hardest (MAE $\approx$ 8.01).

7 Conclusion

This paper presents the first systematic evaluation of six forecasting approaches on the Slovenian national air quality monitoring network (21 stations, 296,100 hourly observations). We identified and resolved a critical lag-feature flaw prevalent in applied ML-based air quality forecasting. Our proposed STL-augmented TCN-BiLSTM achieves a **29% MAE reduction** (2.27 vs. 3.20 μg/m^3) and a **30% RMSE reduction** (3.63 vs. 5.20) over the naive persistence baseline at the 1-hour horizon on station E421.

The multi-station evaluation reveals significant station-level heterogeneity, with E802 (MAE = 8.01) requiring spatial modelling to overcome its episodic pollution dynamics. These results provide a solid methodological baseline for future research on this dataset, including spatio-temporal GNN models, source apportionment via clustering, and integration with a live ARSO data feed for real-time deployment.

Data Availability. The dataset is publicly available on Mendeley Data under CC BY 4.0: `https://data.mendeley.com/datasets/p86hnykz3d/2` [3]. All analysis code and the AQI_Pipeline notebook are available at `https://github.com/vaibhav7766/AQI-data-analysis`.

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